Astronomy A3: Solutions

Relativity and Black Holes · MAIS PAC

Problem 1.

(a) For Earth, $\mathbf{R_s} = 2(6.67 \times 10^{-11})(5.97 \times 10^{24})/(3.00 \times 10^{8})^2 \approx 8.85 \times 10^{-3}$ m = **8.9 mm**—the size of a small glass marble! For Sagittarius A*, $\mathbf{R_s} \approx 2.95$ km $\times (4.15 \times 10^{6}) \approx$ **1.22×10^{7} km ≈ 0.082 AU** (comfortably inside Mercury's 0.39 AU orbit).

(b) Nothing at all happens to Earth's orbit! Outside any spherical mass **M**, the gravitational field depends solely on total mass **M** and distance **r** = 1 AU (Birkhoff's theorem, the relativistic cousin of Newton's shell theorem). Black holes are not cosmic vacuum cleaners: Earth would continue orbiting once a year at 29.8 km/s, though sunlight would stop arriving 8.3 minutes later.

Problem 2.

(a) Substituting $\mathbf{R_s} = 2\mathbf{GM}/\mathbf{c}^2$ into both definitions gives

$$\bar{\rho}_\mathbf{s} = (3\mathbf{c}^6)/(32\pi\,\mathbf{G}^3\mathbf{M}^2) \propto \mathbf{M}^{-2},\quad \Delta\mathbf{a}(\mathbf{R_s}) = (2\mathbf{GM}\ell)/(\mathbf{R_s}^3) = (\mathbf{c}^6\ell)/(4\,\mathbf{G}^2\mathbf{M}^2) \propto \mathbf{M}^{-2}. \tag{1}$$

(b) For $\mathbf{M} = 10\,\mathbf{M}_\odot = 1.99 \times 10^{31}$ kg and $\ell = 1.8$ m, $\Delta\mathbf{a}(\mathbf{R_s}) \approx 7.4 \times 10^{7}$ m/s^2 ≈ **7.5×10^{6} $\mathbf{g_0}$**—tearing steel or astronauts apart ("spaghettification") long before reaching the horizon. For $\mathbf{M} = 10^{8}\,\mathbf{M}_\odot$ (10^{7} times heavier, so 10^{14} times weaker!), $\Delta\mathbf{a}(\mathbf{R_s}) \approx 7.4 \times 10^{-7}$ m/s^2 ≈ **7.5×10^{-8} $\mathbf{g_0}$**, completely unnoticeable as you drift across the event horizon! Even more surprisingly, $\bar{\rho}_\mathbf{s}$ for a $10^{8}\,\mathbf{M}_\odot$ black hole is about **1.85 kg/m^3**—only slightly denser than room-temperature air.

Problem 3.

(a) Evaluating $\mathbf{E} = -\mathbf{GMm}/(2\mathbf{r})$ at $\mathbf{r} = 6\mathbf{GM}/\mathbf{c}^2$ cancels **GM** immediately:

$$|\Delta\mathbf{E}| = (\mathbf{GMm})/(2\,(6\mathbf{GM}/\mathbf{c}^2)) = (1)/(12)\,\mathbf{mc}^2 \approx 0.083\,\mathbf{mc}^2\ (8.3\%), \tag{2}$$

roughly an order of magnitude more efficient than hydrogen fusion (0.7%)!

(b) From $\mathbf{L} = \eta\,(\mathbf{dM}/\mathbf{dt})\mathbf{c}^2$ with $\mathbf{L} = 10^{13} \times 3.83 \times 10^{26}$ W = 3.83×10^{39} W and $\eta = 0.10$:

$$(\mathbf{dM})/(\mathbf{dt}) = (3.83 \times 10^{39}\ \text{W})/(0.10\,(3.00 \times 10^{8}\ \text{m/s})^2) \approx 4.26 \times 10^{23}\ \text{kg/s} \approx 6.8\,\mathbf{M}_\odot/\text{yr}. \tag{3}$$

Swallowing fewer than seven solar masses per year allows a single compact nucleus smaller than our Solar System to outshine a hundred Milky Way galaxies combined.

Problem 4.

(a) Since 1″ at 1 pc subtends 1 AU by definition of the parsec, 0.125″ at 8,200 pc corresponds to physical semi-major axis $\mathbf{a} = 0.125 \times 8{,}200 = \mathbf{1{,}025\ AU}$. Applying Kepler's third law in solar units ($\mathbf{M}/\mathbf{M}_\odot = (\mathbf{a}/\mathrm{AU})^3 / (\mathbf{P}/\mathrm{yr})^2$):

$$(\mathbf{M})/(\mathbf{M}_\odot) = ((1{,}025)^3)/((16.05)^2) \approx (1.077 \times 10^9)/(257.6) \approx 4.18 \times 10^6\ \mathbf{M}_\odot. \tag{4}$$

(b) At pericenter, S2 sweeps within $\mathbf{r_p} = \mathbf{a}(1 - \mathbf{e}) = 1{,}025(1 - 0.885) \approx \mathbf{118\ AU}$ of Sagittarius A*, proving that over four million solar masses lie packed inside a region barely wider than our Solar System (work recognized by the 2020 Nobel Prize in Physics). Combining the vis-viva equation $\mathbf{v_p}^2 = \mathbf{GM}(2/\mathbf{r_p} - 1/\mathbf{a}) = (\mathbf{GM}/\mathbf{a})(1 + \mathbf{e})/(1 - \mathbf{e})$ with Kepler's third law $\mathbf{GM}/\mathbf{a} = (2\pi\mathbf{a}/\mathbf{P})^2$ gives

$$\mathbf{v_p} = (2\pi\mathbf{a})/(\mathbf{P})\ \sqrt{((1 + \mathbf{e})/(1 - \mathbf{e}))} = (2\pi\ (1{,}025 \times 1.496 \times 10^{11}\ \mathrm{m}))/(16.05 \times 3.156 \times 10^7\ \mathrm{s})\ \sqrt{((1.885)/(0.115))} \approx 7.7 \times 10^6\ \mathrm{m/s} \approx 0.026\mathbf{c}\ (2.6\%\ \text{of}\ \mathbf{c}). \tag{5}$$

At 2.6% of the speed of light, relativistic gravitational redshift and Schwarzschild pericenter precession become directly measurable in S2's starlight!